

Coordination in Dynamic Teams: Investigating a Learning–Productivity Trade-Off

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Abstract. The rapid growth in the use of teams and multiple team membership has led more organizations to structure work around *dynamic teams*, characterized by short lifespans and fluid membership boundaries. This approach to organizing work makes it difficult for teams to work efficiently or to support the learning and growth of members. Given the importance of productivity and learning, we ask how these processes can be supported in dynamic teams. We do so in a randomized controlled field experiment conducted with 91 teams in a teaching hospital, a context in which physician trainees must learn while providing patient care in dynamic teams of care providers. We randomly assigned physician teams, or “core” teams, to launch their work with an intervention focused either on internal coordination among the core team members or on external coordination with a changing cast of external contributors such as nurses and specialists. We measured resulting levels of internal and external coordination over teams’ week-long lifespans and observed that external coordination was associated with improved team efficiency, whereas internal coordination was associated with improved individual learning. Post hoc exploratory analysis suggested that individual learning was highest when teams achieved high levels of both internal and external coordination, and it was positively correlated with improvement in the team’s patients’ average length of hospital stay. We discuss the implications of the demonstrated causal effects of team launches, along with our exploratory findings, for the management of dynamic teams and the opportunities to mitigate potential trade-offs between learning and productivity.

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There has been a dramatic rise in the use of teams as a primary structure for organizing work in many settings (Weidmann and Deming 2021). Moreover, the number of teams individuals contribute to simultaneously has also risen sharply as organizations leverage unique knowledge and skills across more areas (Wuchty et al. 2007, O’Leary et al. 2011, National Research Council of the National Academies 2015, Haeussler and Sauer-mann 2020). Given the accelerated pace of change in their environments, these teams are also increasingly *dynamic teams*, characterized by short lifespans and fluid membership boundaries (Mayo 2022), such that they take different forms in response to shifting demands

(Edmondson 2012, Tannenbaum et al. 2012, Wageman et al. 2012, Mortensen and Haas 2018).

The departure from more traditional team structures that were bounded, stable, and interdependent (Hackman 1987) poses opportunities for agility, but also significant challenges. First, dynamic team structures can hamper team productivity (O’Leary et al. 2011, Mayo 2022, Rishani et al. 2024). Second, they can create difficulties for team and individual learning. With rapid advances in technology and the associated growth in information and new knowledge, working professionals need to continuously update their skills. Consequently, learning has become a growing priority in

many work settings (Bersin and Zao-Sanders 2019), in large part because of its relation to organizational effectiveness (Argote and Miron-Spektor 2011). Team membership stability provides important support for team learning as well as individual learning, suggesting that dynamic teaming is likely to inhibit both (Edmondson et al. 2001, Harvey et al. 2022). Moreover, investment in one's own learning and the development of others takes time and attentional resources, and this could reduce short-term productivity (Sobrero and Roberts 2001, Singer and Edmondson 2008, Dennis et al. 2014, Rabino-witz et al. 2016). Indeed, dynamic teamwork could exacerbate the documented tensions between learning and productivity (Singer and Edmondson 2008) as commanding the shared attention needed to accomplish either in these settings is extremely difficult.

A natural solution to the problems created by the absence of structure in teams could be to create new structure, but exactly what kinds of structure would support both learning and productivity in dynamic teams is not clear. Extant research on team productivity and performance demonstrates an important role of external coordination—with constituencies outside of the team boundary—to both provide and receive information and facilitate task coordination by creating ties to others in the team's environment (Ancona and Caldwell 1988, Choi 2002, Mayo 2022). By contrast, research on learning in teams has highlighted the importance of internally focused team process and coordination, such as practices that facilitate knowledge sharing and integration (for a review, see Edmondson et al. 2007). In an ideal situation, a team would attempt to address both internal and external coordination. However, much as some extant work highlights the potential tension between learning and productivity in teams, as both compete for limited time and attention (Singer and Edmondson 2008), research also points to potential tensions between internal and external coordination, with teams often trading off between the two (Ancona and Caldwell 1988, 1992; Ancona 1990). Moreover, given the short lifespans and fluid boundaries of dynamic teams, there is reason to look to a growing body of research that points to the advantages of short, focused interventions, such as the Safe Surgery Checklist or daily health-care huddles, which are successful in highly time-pressured environments (Kilpatrick et al. 2020, Murphy 2023). Taken together, it is essential to identify the minimally viable set of structures that will best support dynamic team functioning.

To address the question of what minimally viable set of structures would best support dynamic teaming, we conducted a randomized controlled experiment with 91 general pediatric inpatient teams to test two different short, targeted interventions, both delivered to a team's core members (i.e., those contributors who are relatively more stable and central to decision making; Ancona and

Bresman 2007, Humphrey et al. 2009). One intervention focused on internal coordination of tasks and role responsibilities among the core team members, and the other focused on coordination of the core team with outside contributors. Core teams received one intervention or the other as they convened to begin working together, and we examined the effects on coordination behaviors and resulting team productivity and individual learning. We observed that each intervention increased the targeted coordination behaviors, as expected. Our results further demonstrated that external coordination was associated with improved team productivity based on the percentage of morning discharges. Additionally, teams' internal coordination was strongly related to individual learning, though an additional post hoc analysis revealed that this was qualified by an interaction effect: individual learning was highest when teams achieved both high internal *and* external coordination. Finally, in post hoc exploratory analyses, we observed that the individual learning of team members was associated with their patients' adjusted length of stay, demonstrating a potential for synergies between individual learning and team productivity. Additional post hoc exploration also suggested that core teams receiving the internally focused intervention were slightly more likely to achieve high levels of both internal and external coordination which produced the synergistic learning and productivity outcomes.

In reporting this field experiment, this paper makes several theoretical and practical contributions. This research adds to the research on dynamic teams—which heretofore has been largely theoretical (Tannenbaum et al. 2012, Wageman et al. 2012, Edmondson and Harvey 2018, Mortensen and Haas 2018), or correlational or qualitative in nature (e.g., Majchrzak et al. 2007, Mayo 2022)—in that it supplies vital causal evidence of the impact of focusing attention squarely on internal versus external coordination in a real-world setting permitting direct comparisons, revealing some unique properties that result from each. By distinguishing between the role of external coordination for team efficiency, and the role of internal coordination for member learning, we also contribute to the emerging theoretical understanding of dynamic teams. At the same time, we contribute to research exploring the relationship between learning and productivity by demonstrating the potential for synergy, where individual learning and team productivity can be mutually reinforcing, and our post hoc explorations suggest implications for theory development and avenues for future research to capitalize on this. In supplying this evidence, our paper also makes a significant practical contribution to the management literature on managing dynamic teams by demonstrating the efficacy of simple, low-cost interventions to enhance learning and productivity in a challenging teamwork environment. Finally, this work

integrates and builds on extant research on team starts (Ericksen and Dyer 2004, Mathieu and Rapp 2009, Woolley 2009, Ginnett 2010, Hackman 2011) and team huddles (e.g., Rodriguez et al. 2015, Franklin et al. 2020). Whereas extant work on team starts and huddles presumes the ability to convene a full team (Jain et al. 2015, Rodriguez et al. 2015) and typically focuses on internal team structures and processes (Mathieu and Rapp 2009), we extend this work to dynamic teams and fold in consideration of external coordination and the differential implications for team productivity and individual learning.

Theoretical Background

Coordination in Dynamic Teams

In order to adapt rapidly to volatile environments, organizations are increasingly using “just-in-time” staffing models (Capelli 2020) and temporary teams (Faraj and Xiao 2006, Valentine 2018, Kim et al. 2023). A consequence of these trends is an increase in dynamic teams: characterized by short lifespans and fluid membership boundaries, resulting from fluctuating membership and unclear boundaries as members drift in and out of the team’s work (Mayo 2022).

As the membership boundary around teams has become increasingly fuzzy over the last few decades, researchers have introduced ways to conceptualize or clarify members based on their level of involvement in a particular team. For instance, the distinction between “core” and “noncore” contributors has gained currency as a means of distinguishing different roles (Ancona et al. 2002, Humphrey et al. 2009, Cummings and Pletcher 2011), and it parallels concepts in networks research that distinguish core and periphery members (Long and Siau 2007, Garcia et al. 2021). Core team members are more central to decision making and exhibit more stability in their collective work relative to the peripheral members whose contributions may be critical but who lack decision-making authority and stability (Ancona and Bresman 2007, Humphrey et al. 2009). Peripheral members are often involved to offer a unique contribution of knowledge or supply additional resources to support the core team’s work (Cummings and Pletcher 2011).

One of the key processes widely acknowledged as fundamental to teamwork is coordination (Marks et al. 2001), or the “temporally unfolding and contextualized process of input regulation and interaction articulation to realize a collective performance” (Faraj and Xiao 2006, p. 1157; as cited also in Okhuysen and Bechky 2009). Because the core group comprises only a subset of the broader network of team members involved in carrying out work, they often must coordinate work across professional, unit, and departmental boundaries or engage in “boundary work” (Chreim et al. 2013, Meier 2015,

Langley et al. 2019), to exchange information, delegate, and sequence events with periphery members. Accomplishing coordination either within the core group (internal coordination) or with those periphery members (external coordination) is nontrivial. The temporary nature of the team limits core team members’ ability to learn how to work together. At the same time, coordination with periphery members in dynamic teams is difficult for multiple reasons. First, dynamic teams often cannot know, *a priori*, all of the external members who might contribute throughout the team’s lifespan, as the composition of that group constantly changes either as a result of shift changes or the evolving demands of the work (Humphrey et al. 2009). This makes it difficult to clearly articulate a team boundary at a team start that could direct attention and therefore guide coordination. Second, integrating inputs from outside a core group is difficult given that the boundary differentiating core and periphery often aligns with other differences in perspectives or training (Hall 2005). Indeed, such cross-boundary work does not always come naturally, with a tendency for members to instead focus on the demands of their personal roles (Edmondson 2012). Finally, when periphery members are scattered throughout an organization, working across boundaries is even more complicated, as they tend to be “out of sight, out of mind” (Kiesler and Cummings 2002, p. 63).

Traditional approaches to structuring coordination often involve using clearly distinguished roles and standardized routines to enable members’ inputs to fit together. Although some aspects of a dynamic team’s work can be facilitated using such structures, imposing standard operating procedures may result in a lack of adaptation to the dynamism inherent in the team and its environment, or misaligned expectations as some members adjust as needed while others do not (Lifshitz-Assaf et al. 2021). Consequently, a more successful approach may lie in creating a minimal and flexible structure (Lifshitz-Assaf et al. 2021). Because gathering the entire set of contributors to a dynamic team is infeasible, intervening with only the core members of a dynamic team, who typically interact with each other more frequently, is much more achievable. Indeed, a qualitative study with dynamic teams suggests this may be fruitful (Mayo 2022). Further, building on extant work on the value of team launches (Marks et al. 2000, Hackman and Wageman 2005) and sufficiently brief team huddles (Franklin et al. 2020, Kilpatrick et al. 2020), and folding in consideration of periphery members in addition to the core, we suggest that an intervention with the core group could direct attention to the (flexible) core-team roles and tasks and thereby support internal coordination, or direct attention to the (flexible) set of contributing periphery members and thereby support external coordination. Building from these ideas,

below we further develop our theoretical arguments and predictions related to the benefits of external and internal coordination with respect to team productivity and individual learning.

Coordination and Productivity

Team coordination can take a number of forms, depending on the team's task and environment, and here we start with a focus on external coordination. Dynamic teams arise more often than not in response to rapidly changing environments, where they must manage dynamic interdependencies (in terms of task, goal, and/or knowledge interdependence; Raveendran et al. 2020) that span professional and functional boundaries, such that external coordination with individuals outside of the core team could allow for adaptively leveraging widely distributed expertise and resources (Edmondson 2012, Tannenbaum et al. 2012, Wageman et al. 2012, Mortensen and Haas 2018). Indeed, they are characterized by environments in which there is some relevant knowledge or capability held by the periphery.

Although the external coordination that a core team carries out with periphery contributors cannot be specified *a priori* and is not easy to achieve, we argue that it has many benefits for team efficiency and productivity. First, dynamic teams are commonly functioning within a network of interdependencies with external contributors that are not identifiable in advance, as those interdependencies tend to be dynamic. For instance, in medicine, the need for a physician to coordinate external information (e.g., gathering input from a nurse or specialist) related to clinical reasoning and care planning evolves over time (Kim 2020, Mayo 2022). Similarly, in product development teams, external coordination can lead to new insights, but it is not obvious where the valuable insights may reside when the team begins its work (Cummings and Pletcher 2011). Second, the degree to which those external interdependencies need to be managed can vary, as research has long acknowledged the costs of cross-boundary coordination (Cummings and Kiesler 2005). Indeed, dynamic teams can incur substantial costs from high levels of external coordination, which, given limited resources, could undercut other necessary internal coordination, leading some to call for only moderate external coordination (Ancona and Caldwell 1988, Ancona 1990, Ancona and Bresman 2007, Ziegert et al. 2022).

Nevertheless, when a team has interdependencies with others in its environment, particularly in environments lacking hierarchical or centralized structures to manage coordination, engaging in external coordination is essential for productivity (Ancona et al. 2002), in part because contributors provide essential information and facilitate critical steps in the execution of tasks (Andersen 2015). For example, in hospital settings, a core physician team is interdependent with multiple other

professions and disciplines in delivering patient care, including nurses, pharmacists, therapists, and other medical specialists. In that setting, when a physician team decides on medication treatment for a patient, that core physician team typically completes a medication order, and subsequently, periphery members such as a pharmacist and a nurse must complete specific steps to complete the process of administering the medication. Core teams that coordinate with the periphery allow the periphery to offer valuable input and ensure the periphery is aware of the next steps, which reduces the need for backtracking (e.g., because the nurse can share relevant information about patient medication preferences before the order is placed) and enables the periphery to execute more efficiently (e.g., because the periphery is aware that an order was placed). This coordination enhances efficiency overall, as demonstrated by extant research showing physicians' effective coordination with pharmacists results in shorter hospital stays for patients (Chisholm-Burns et al. 2010). By contrast, if a core team does not coordinate well with periphery members, it is likely to lack essential input and face an overall delay in execution. The impact of core-periphery coordination has also been demonstrated in industries beyond healthcare, where better coordination enables the core to make use of relevant knowledge and expertise to not only achieve higher productivity but also be more innovative (Cummings and Pletcher 2011). In short, when periphery members have relevant expertise and can assist with task execution, we expect that a core team's external coordination with periphery members—as distinct from the internal coordination that is emphasized by many team interventions (e.g., see Mathieu and Rapp 2009)—will enhance team productivity.

Furthermore, we contend that the level of internal coordination will not directly affect team productivity in the majority of contexts deploying dynamic teams for organizing work. Dynamic teams are often both a result and a driver of interdependence *across* teams in their broader context (O'Leary et al. 2012, Larson et al. 2023), as with high levels of knowledge, goal, and task interdependence across a multiteam system (Raveendran et al. 2020). In dynamic teams, the coordination of the core team with others outside the team is, more often than not, essential, as argued above. High levels of coordination within the core team could also benefit productivity in that this could allow for backup behaviors and mutual adjustment within the core (Marks et al. 2001), and this could enable the team's ability to adapt as new demands arise. However, it could be that the core team opts to divide its work in a way that demands lower levels of, if any, ongoing internal coordination (e.g., with interdependence that emerges as pooled rather than reciprocal; Thompson 1967). Such a choice could be effective in that it could expedite decision speed

(Hollenbeck et al. 2011) within the core and free attention for external coordination, which may be of particular importance given that dynamic teams focusing too much on their internal coordination risk becoming too insular and overlooking essential inputs from their environment (Ancona 1990). Taken together, we argue that for dynamic teams working in their typical environment, external coordination will be a strong driver of team productivity, but internal coordination will not.

Hypothesis 1. *A dynamic team's level of external coordination will predict the team's productivity.*

Coordination and Learning

Although we anticipate that, in a dynamic team, the core team's external coordination will be a key driver of productivity, we anticipate internal coordination will be a key driver of learning among core members. Here, we build on definitions of learning as a change in one's set of knowledge and potential behaviors as a function of experience—an approach consistent with cognitive perspectives across the study of individuals (Anderson 1993, Walsh and Anderson 2012), groups (Wilson et al. 2007, Argote 2013), and organizations (Huber 1991, Argote and Miron-Spektor 2011). We draw a distinction between our focus on individual learning of knowledge and research on team learning. In the latter, research tends to focus on the behaviors that could change collective knowledge (Edmondson 1999, Edmondson et al. 2007, Wiese et al. 2022) while inferring that such collective (and individual) learning has occurred from performance gains (Wiese et al. 2022). Further, although team learning has been shown to be important to team outcomes in a broad range of environments, in dynamic teams, “the concept of team viability seems quaint” (Maloney et al. 2019, p. 275), such that learning to work with any specific team may be a moot point. Instead, organizations using dynamic team structures increasingly focus on supporting individual learning about teamwork skills to enable better team effectiveness (Hughes et al. 2016) as well as supporting individuals to develop a robust and diverse range of skills to enhance team ambidexterity and adaptation (Dean 2021).

We have long known that direct experience (Thorndike 1898) and observation of others (Bandura 1971, Gioia and Manz 2011) facilitate learning. However, recent work elaborates on this foundation to demonstrate that mere exposure to novel information is not sufficient for learning. Instead, rather than being solely an intrapersonal process, individuals can engage in a process of coactive vicarious learning, involving a relational process of “coconstructed, interpersonal learning that occurs through discursive interactions” with others (Myers 2018, p. 610). This is consistent with classic constructivist theory originating in developmental psychology, wherein children (and adults) acquire new

knowledge by assimilating and accommodating new or contradictory information through interaction with others in their environment (Inhelder and Piaget 1958). Building on these seminal ideas, a growing body of evidence demonstrates that in addition to learning from experts, learners can also benefit greatly from collective processing of new information with peers operating at the same level of competence (Myers 2021), as demonstrated by the benefits of peer review and team-based learning in educational environments (Cho and Schunn 2007, Michaelsen and Sweet 2011). Studies in organizational settings demonstrate similar effects of peer coaching, such as when members in interdependent teams can seek and give each other advice to solve problems (Wageman 2001). Consequently, an accumulating body of evidence demonstrates that more coordinated, cooperative, and interdependent work in groups is associated not just with improved team learning (Edmondson 1996, McGrath and Argote 2004, Woolley and Aggarwal 2020) but also with individual learning (Johnson and Johnson 1975, Yager et al. 1986, Wageman and Gordon 2005).

There are, of course, benefits to being exposed to a wealth of new information via contacts with others outside one's own core team, but such external learning activities more readily result in improved capability when they are coupled with the capacity to absorb it (Nemanich et al. 2010, Venkataramani and Tang 2024). Absorptive capacity reflects an ability to understand and integrate other's expertise (Cohen and Levinthal 1990). Though often studied at the firm level, absorptive capacity has more recently been examined at the team (Venkataramani and Tang 2024) and even the individual level of analysis (Siachou and Gkorezis 2014) in response to calls for a better understanding of the micro-foundations of the construct (Volberda et al. 2010, Tian and Soo 2018). At the team level, the idea that internal coordination creates a foundation for absorptive capacity such that teams can better learn from external coordination is supported by research demonstrating that teams can perform better as a result of external learning activities (the behaviors theorized to yield learning, as noted above) only if they also engage in more internal learning activities, underscoring the essential role of processing new knowledge with coworkers (Bresman 2010, Myers 2021). Additionally, we note that research examining a team's ability to capitalize on external inputs has done so by defining absorptive capacity in terms of the team's ability to form a shared understanding and then demonstrating that this internal ability was associated with the ability to apply inputs from outside the team (Nemanich et al. 2010). Critically, research at the individual level is also consistent with this pattern. For instance, research on the correlates of individual absorptive capacity has identified internal motivation and personal empowerment as two important influences,

which are two internal states that are highly influenced by the frequent interactions employees have with coworkers, much like dynamic team members would have if they were engaged in high levels of internal coordination (Conger and Kanungo 1988). By contrast, external coordination might provide opportunities to learn from diverse new information—at least for those who engage in the external coordination—but little to no opportunity for the collective processing essential for individual learning, either by those initially exposed to the external inputs or others who never encounter it. Taken together, the extant literature suggests that whereas external coordination could provide novel inputs and ideas, internal team coordination is vital to fostering the intermember processes that underlie individual learning.

Hypothesis 2. *A dynamic team's level of internal coordination will significantly predict individual member learning.*

Learning and Productivity

Observations from a handful of studies point to a potential trade-off between the level of learning that occurs in a collaborative setting and the team's efficiency or productivity (Sobrero and Roberts 2001, Singer and Edmondson 2008), based primarily on the logic that both require time, which is a zero-sum resource. For instance, research on the benefits of supplier–manufacturer collaboration in new product development has demonstrated that learning by one of the collaborators—the manufacturer—can detract from the short-term efficiency of the collaboration's development process overall (Sobrero and Roberts 2001). A similar tension is also present in team-based settings where individual learning is important, such as in academic teaching hospitals. There, dedicating time to the education of medical residents to support individual learning is expected to slow the team's provision of patient care (e.g., Dennis et al. 2014, Rabinowitz et al. 2016). Thus, studies that have explicitly examined the relationship between learning and productivity at different levels of analysis have observed similar trade-offs, typically attributed to the zero-sum nature of time as the primary explanation. Other work recognizes the initial trade-off but suggests a delayed benefit of learning—a “worse-before-better” effect—noting that learning involves experimentation that takes time to conduct and may come with some errors, and thus the benefits may take time to emerge (Singer and Edmondson 2008, p. 39; Nembhard and Tucker 2011, p. 918). For instance, the literature on learning curves—from the study of individuals to teams to organizations—highlights this nuanced tension between learning and productivity, where gains in experience (thought to go hand in hand with learning) can enhance performance (Argote and Epplé 1990); though, there may be an initial decrease in

productivity before gains are realized (Musaji et al. 2020). While recognizing that cross-level effects are not always isomorphic, we take the robustness of these effects across levels as compelling, and we both build from and map the zero-sum and worse-before-better perspectives to dynamic teams. In doing so, we note that the difficulties of carrying out the collective processing necessary for learning in such a complex setting could exacerbate any potential trade-off between learning and productivity derived from having scarce attentional resources, and the short team lifespans could impede opportunities for learning to yield performance gains.

However, there may be conditions under which the trade-off assumptions do not hold, and instead there could be an opportunity for even near-term synergistic gains in dynamic teams. This could occur because of members' increased competency from learning being translated more immediately into productivity, or by exposure to better ways of doing things fueling both individual learning and, thus, group productivity. For instance, Sarin and McDermott (2003, p. 717) examined team learning in terms of “how much *members* had learned while conducting the project” (emphasis added), and found that their measure of learning predicted higher productivity in product development teams as reflected in quicker speed to market. Similarly, Kostopoulos and Bozionelos (2011) showed that *members'* reports of their own *new skill development* while working on a team project predicted manager ratings of team productivity. Of note, these studies emphasize individual learning that is achieved while working with others, and thus their findings suggest that collective and informal learning (rather than more didactic or formally reflective learning; e.g., see Nembhard and Tucker 2011) may be key to unlocking near-term synergies. For instance, research suggests that collective work to address challenges and errors can support individual learning (Edmondson et al. 2007, Edmondson and Nembhard 2009) as well as unit performance improvement (Edmondson 1996), whereas a qualitative study of nurses suggests that when a nurse addresses a failure independently, others do not have a chance to also learn from that failure, and even if the nurse learns, this can limit the ability for individual learning to support unit-level performance (Tucker and Edmondson 2003). Moreover, collective processing not only supports an individual's learning in terms of gaining new knowledge but is also essential to gaining the ability to apply it (Myers 2018). Thus, there appear to be opportunities to find synergies between individual learning and productivity, but how best to plant the seeds for them to occur is not entirely clear.

In sum, we take these findings to suggest that—at least under certain conditions—individual learning within a team context can enhance a team's productivity. We stop short here of making formal predictions about the specific conditions moderating the relationship

between learning and productivity, given the mixed evidence in the literature and the complexity of the contingencies surrounding these relationships. Instead, we will present exploratory analyses to see what evidence exists supporting or refuting our intuitions. We then build on those empirical observations in our discussion to consider the implications for theory and future research. We illustrate our conceptual model in Figure 1, distinguishing our evidence-based predictions from the relationships we explore in our post hoc analyses.

Research Setting

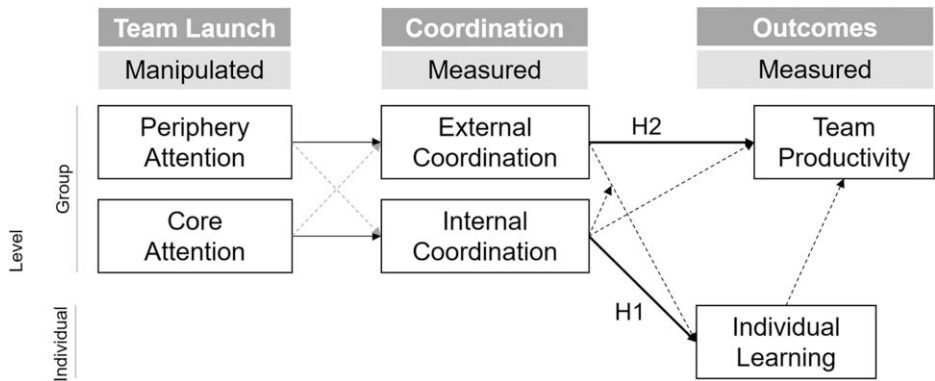
This research took place within the general pediatric inpatient service of a large and free-standing academic children’s hospital. Here, there was an apprenticeship-like model to medical education such that physician teams were tasked with educating trainees, as well as determining and executing care for a patient and deciding when that patient was ready to leave the hospital. These physician teams were assigned to a shared panel of patients over which they had decision-making authority. These teams conducted their work with assistance and input from nurses, specialists, pharmacists, social workers, care coordinators, patients and families, etc.

Following extant research, we differentiated the core team from periphery members based on (1) decision-making authority and (2) consistency of engagement with the team (Ancona and Bresman 2007, Humphrey et al. 2009). In our setting, core members included only the physician team: typically one attending physician (supervising physician), one senior resident (second or third year of residency), two interns (first year of residency), and two to three medical students. This core team had decision-making authority for their patients, and core members worked together for one week at a time, sharing all of their patients. Although sometimes the core members have higher status in the organization than periphery members, this is not always the case. For

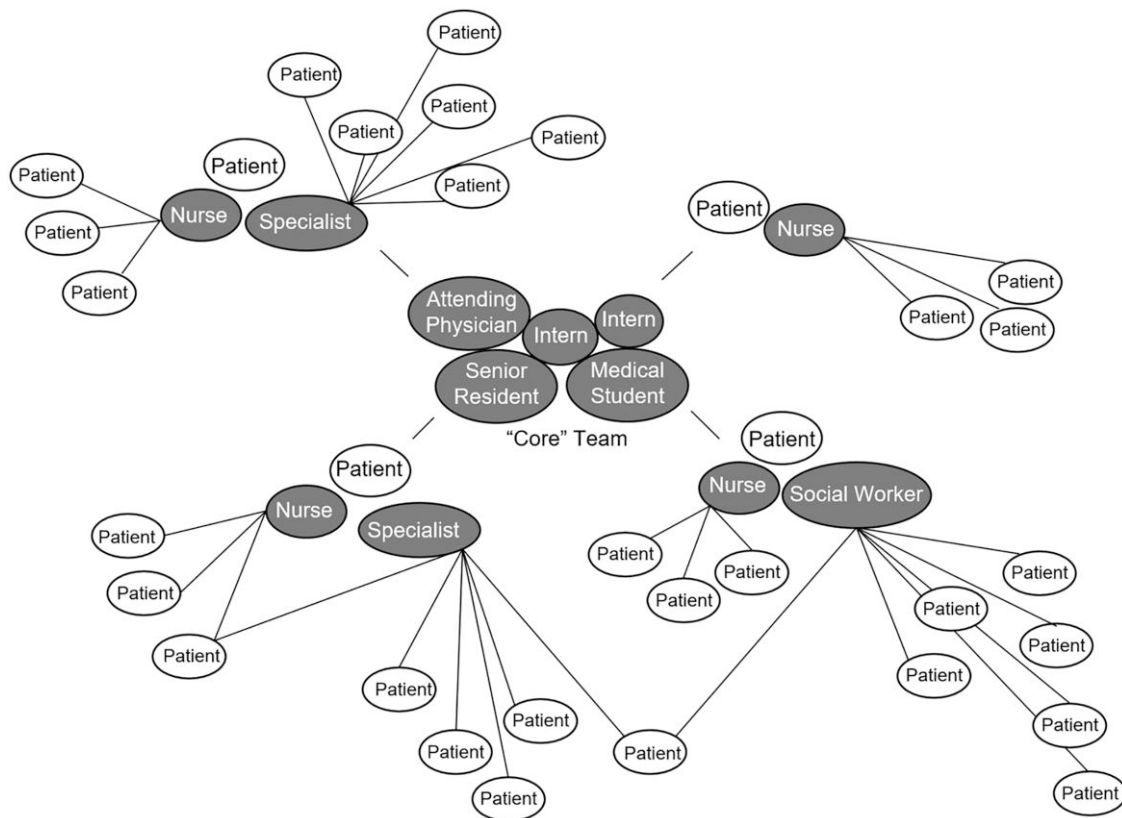
instance, nurses are considered periphery members, not because they are considered lower status than physicians, but because they did not have the same decision-making authority. Further, even though they might spend more time caring for a patient than the physicians on the core team, they could rotate on and off service as frequently as daily and shared only a small (often just one) of the core team’s overall set of patients. By that same token, medical specialists, who are often considered higher status than primary care physicians, are also periphery members in this setting, as they also did not have ultimate decision-making authority and made narrower and more temporary contributions to a case than the core team.

For each patient cared for by a core team, there was a different set of periphery members such that each patient’s care team was typically unique in its composition and also experienced changing membership resulting from shift changes or developments in the patient’s condition. (See Figure 2 for a stylized illustration of care team structure in this setting.) Because of the shifting composition of the periphery roles involved in each case, there was no standard operating procedure for determining accountability for specific tasks. For instance, even though a nurse was assigned to every patient, and there was a range of activities for which the physician team *could* include the nurse or a specialist, not all of these tasks necessarily had to involve periphery contributors. Instead, these connections could vary as a function of the patient details as well as the idiosyncratic preferences of the core team. As is common in medical settings, much of the planning of patient care occurred during morning rounds, when each core team discussed and made decisions about the care for each of its assigned patients (Ervin et al. 2018). At this hospital, core teams were instructed to include patient families and nurses during morning rounds; however, preliminary data collected in this setting demonstrated that the

Figure 1. Conceptual Model



Notes. When predicting coordination behaviors, gray dashed lines are controlled for, such that we can determine whether each intervention has distinct effects on subsequent coordination behaviors. When predicting outcome measures, the black dashed lines reflect relationships that we explore, but for which extant research evidence is not substantive enough to make a strong prediction. H1, Hypothesis 1; H2, Hypothesis 2.

Figure 2. Illustration of Care Team Structure in the Research Setting

Note. Illustration of one core team (depicted in the center) and its ties to various periphery team members via shared patients, wherein the periphery members (those external to the core) are themselves working on many different patient cases (indicated in white).

degree to which inpatient physician teams included nurses in their morning rounds varied from 30% to 85%, suggesting significant room for improvement in core team external coordination.

Methods

Sample and Design

The overall sample included 243 unique core team members (21 attending physicians, 222 trainees) from the general pediatric in-patient service and 33 unique periphery members (nurses) who shared their patients. Together, over the course of the study, these individuals composed 91 teams.¹ Each week four new core teams were constituted from the same pool of employees. Scheduling for all members was constrained by shift and/or training requirements, leading membership in each core team to be largely random, in that it was not dictated by ability, preferences, or history working together. Because of the reconstitution of each core team each week, we treated the core teams from week to week as separate entities such that the unit of analysis was a core team in a given week, along with the periphery members associated with each patient case. Our

study design included three conditions: a control condition and two distinct interventions.

Interventions. The team launch interventions involved only core team members and were implemented at the core team's start (Monday at 9:00 a.m.), when a core team's senior resident led fellow core members through the assigned intervention, which averaged 10 minutes.²

Intervention materials are provided in Online Appendix B. Building from prior qualitative work (Mayo 2022), we designed two interventions—team launches—that focused either on internal coordination within the core or external coordination with the periphery. Broadly speaking, the interventions were designed to direct the team's attention to specific issues in order to shape their coordination behavior. The “core-attention” team launch directed attention toward internal coordination, specifically to roles and workflows among core team members. The “periphery-attention” team launch directed attention toward external coordination with potential periphery members and relevant workflows. In structuring the interventions, we followed evidence-based change management practices and theory about “wise” interventions

including: (1) motivation for the change, (2) removal of barriers to the change, and (3) the enabling of autonomous work (Walton 2014, Stouten et al. 2018, Walton and Wilson 2018). Consequently, each script has a motivation for why focusing on the selected issues is important, removes a barrier by directing the team to gather contact details for other core or periphery members (as the lack of that information was noted in preliminary observations as a major barrier to coordination), and provides a discussion guide for collective planning on how the core team wanted to address the relevant focal issues. Each discussion guide—the latter element—included the same number of questions, with the core-attention version focused on how members coordinate with each other, and the periphery-attention version focused on who to coordinate with outside the team and how to do so.

Block Random Assignment. The study involved two phases. The first phase (September 2017 to December 2017) included data collection for the control condition ($n = 48$: for 12 weeks, four core teams were observed each week, identified by a color: Blue, Green, Red, and Purple). During the second phase (January 2018 to March 2018), we used block random assignment to assign core teams to one of the two intervention conditions (the “core-attention team launch” or the “periphery-attention team launch” intervention; $n = 24$ per intervention: for twelve weeks, two core teams were assigned to each intervention each week). Block random assignment allowed us to prevent the contamination that would occur if a core member received one intervention in one week, but a different intervention in the next week. This was a risk because members worked on multiple teams across different weeks in the study. Thus, in cooperation with the hospital administration, we divided the pool of physicians staffing the core teams into two blocks (i.e., the Blue/Green teams and the Purple/Red teams). Each block was then randomly assigned to an intervention condition, and staffing assignments during the intervention phase of the study were restricted so that each physician was always assigned to a team within the same block (i.e., physicians were assigned only to the Blue or Green team, or only to the Purple or Red team).

Condition Comparison. When comparing each team launch to the control condition, it is important to note the possibility that from phase one to phase two of data collection the passage of time could have affected the behaviors and performance of core team members. We argue that these effects would likely be a reflection of other environmental conditions—namely, the potential for core team members working later in time to have obtained more experience working in their specific positions on these inpatient teams, and the potential for a specific week to result in a higher patient load, for

example, during flu season. We addressed this by using control variables in our models (e.g., core team members’ experience in their position, the core team’s experience working together, and the core team’s patient load; each is discussed in more detail below).

A key benefit of this study’s design is that, although the control condition was implemented before the team launches, the team launches were implemented concurrently and thus served as controls for each other. This allowed us to assess team launch effects as driven by the team launch substance and not time (e.g., gained experience) or demand effects (e.g., from receiving a team launch of any kind). Further, the simultaneous implementation of team launches reduced the likelihood that core team members participating in a team launch would share details across the two conditions, as all members in all core teams in the general pediatric inpatient service received some form of team launch, compared with if some core teams were receiving some form of a team launch while other core teams received nothing.

Measures

Core and periphery members were surveyed at the end of each week; all survey scales were responded to on a seven-point scale (1 = strongly disagree, 7 = strongly agree) unless otherwise noted. Productivity measures and controls were based on hospital scheduling and electronic medical records.

External Coordination. We obtained our measure of external coordination from periphery members who shared patients with a focal core team in the study. In this setting, we collected our ratings of external coordination from nurses because they were the periphery team members who were most consistently involved in patient care. Nurses deal with different core teams, often a different team for each of their patients, making them an excellent informant about core team external coordination. All nurses were blind to the study design and team launch materials, including the condition assignments of core teams. On Wednesday and Friday each week, we asked the nurses to respond to a short survey about each team they worked with that day from the general pediatric inpatient service.

For our measure of external coordination, drawing on the literature on coordination in organizations (i.e., Okhuysen and Bechky 2009), we included items capturing both the structural and emergent components coordination (i.e. the routine and the emergent action), as the two are orthogonal, and teams engaging in both can be interpreted as engaging in more coordination than teams that do only one or none. The survey included items capturing these two components of their coordination with the core team for each patient. For the structural component, nurses were asked whether the core

team included them on their morning rounds for that patient, and we aggregated responses for each core team to calculate the percentage of patients for which nurses had been included in morning rounds. For the emergent component, we asked the nurse to rate the extent to which “The team included me in the decision making” and “The team valued my input” (scale: 1, strongly disagree, to 7, strongly agree; $\alpha = 0.91$) because coordination resulting in those perceptions could occur during morning rounds or at other times. We had an 83% response rate to these surveys, and an average of 4.9 nurse responses per core team (min = 2, max = 10); we excluded three teams receiving fewer than two responses. The responses to the two emergent coordination items exhibited acceptable levels of internal agreement and between-group variation to support aggregation ($F = 1.48$, $p = 0.005$, $ICC(1) = 0.09$; $ICC(2) = 0.32$; median $r_{WG(2)} = 0.61$, mean $r_{WG(2)} = 0.75$). We z-scored the two separate components of the external coordination measure ((1) percentage of nurses invited to rounds and (2) average of the two Likert-scale survey scores) and averaged them to create an overall external coordination score ($\alpha = 0.83$).³

Internal Coordination. Because the main structural component of internal coordination was morning rounds (which were conducted by all core teams), for our measure of core team *internal coordination* we focused on emergent coordination. This was captured based on core team members’ average response to the item “We tended to run the list after rounds, discussing each patient’s tasks/updates” (participation rate = 90%). This practice, “running the list,” was known by name because it is commonly done at handoffs (as a structural coordination mechanism), but it is also an example of emergent coordination because it involves core team members spontaneously going beyond their structural coordination routine of morning rounds to spontaneously consider open tasks, prioritize them, and delegate. It is also a practice that varies considerably across teams. The level of agreement among members’ responses was sufficient to justify aggregation at the group level (between-group variance was significant: $F = 2.96$, $p < 0.001$; $ICC(1) = 0.22$; $ICC(2) = 0.66$; mean $r_{WG(1)} = 0.81$; median $r_{WG(1)} = 0.86$).

Learning. We assessed individual *learning* in terms of task mastery (Hoegl and Gemuenden 2001) based on responses of core team members in trainee roles (i.e., excluding the attending physician) to the items “I was able to acquire important know-how during this week” and “I learned important lessons from this week” (scale: 1, strongly disagree, to 7, strongly agree; $\alpha = 0.87$). This measure was analyzed at the individual level for hypothesis tests. In exploratory analyses, we examined the relationship between core team members’ average

learning and team productivity (i.e., to predict the group-level outcome of productivity, we aggregate learning to the group level by taking the average).⁴

Productivity. We assessed productivity in two ways. First, as a measure of each core team’s daily task productivity, we calculated *morning discharges*, which is the percentage of each team’s patient discharges that occurred between 6:00 a.m. and 11:00 a.m. The hospital strongly encouraged this because it improved throughput from the hospital’s emergency department, which is often busy and needs to be able to send sick patients into the inpatient unit where beds are finite. As a second and perhaps more comprehensive measure of productivity, we calculated each patient’s severity-adjusted length of stay: length of stay divided by the patient’s All Patient Refined Diagnosis Related Group (APR DRG) weight, a measure of case severity and complexity. We then took the average of a team’s patients scores, resulting in an *average severity-adjusted length of stay* (average ALOS) for a core team’s patients. ALOS scores are commonly used for comparison of productivity across units in hospital care (Gross et al. 1997, Lu et al. 2015), as they capture the efficiency of care while controlling for case severity (as more severe cases usually require longer hospital stays).⁵

Control Variables. We accounted for several potentially confounding variables by including them as controls in our analyses (see Table 1 for a comparison of all control variable means by condition). Core teams across conditions marginally differed in their number of members or *core size* ($F = 3.02$, $p = 0.054$), which can affect the difficulty of coordination. Core teams also differed by condition on *patient load* (the number of patients admitted to the core team during the week; $F = 4.96$, $p = 0.009$), which can also impact coordination processes and difficulty. *Core experience working together* can also be a significant advantage for unit performance (Reagans et al. 2005). Following prior work, we calculated the number of weeks during which each dyad within the core team had previously worked together during the year, then aggregated to the team level. Core teams did not differ by condition in their experience working together, nor in the *average case severity* of patients,⁶ but we controlled for these variables given the potential effect on outcomes of interest based on extant work.

To control for *senior resident experience* and *average intern experience*, we used staffing records starting with the beginning of the training year (over two months prior to our study launch) and calculated the number of weeks senior residents and interns had served in their roles on the general pediatric inpatient service prior to and including the week of observation. We also control for *attending experience* based on the number of years of experience of each team’s attending physicians (i.e.,

Table 1. Comparison of Control Variable Means by Condition

Variable	Control	Core-attention team launch	Periphery-attention team launch
Core size	7.00	6.61	6.36
Attending experience (years)	16.06	14.52	17.18
Senior resident experience (weeks)	1.63 ^a	3.39 ^b	2.31 ^a
Average intern experience (weeks)	2.24 ^a	3.29 ^b	3.00 ^b
Core experience working together (weeks)	1.17	1.19	1.23
Patient load (admitted patients)	12.65 ^a	17.04 ^b	15.04 ^c
Average case severity	3.91	3.96	4.29

Notes. ^{a,b,c}For each variable, distinct letters indicate a significant difference across condition, $p < 0.05$. Lack of a letter indicates there is not a significant difference across conditions.

years practicing since board certification). Across the three conditions, teams did not differ in the experience of attending physicians; however, they did differ in senior resident experience ($F = 7.69$, $p < 0.001$) and intern experience ($F = 7.71$, $p < 0.001$; see Table 1). Finally, we also controlled for the week in which each core team observation occurred, as time can also serve as a general indicator of the amount and type of work a core team will face.

Results

Descriptive statistics and correlations are shown in Table 2. Because core team members could be assigned to different core teams over the course of the data collection period, there is a lack of independence of our observations of different teams, which violates a key assumption of ordinary least squares models. Multiple membership models (Browne et al. 2001) account for this lack of independence. We therefore implemented multiple membership models using the software R and MLwiN, and the R package R2MLwiN (Zhang et al. 2016), when predicting variables assessed at the group level. To predict individual learning, we implemented hierarchical linear models using the R package lme4 and the function lmer to account for the nesting of the learning measure in both individuals (i.e., some individuals

appear in the data more than once) and in teams. Model results are reported in Tables 3–6.

Effects of Team Launches on Internal and External Coordination

To verify whether our randomly assigned treatment, the team launches, had the intended effects on core teams' coordination, we implemented multiple membership models predicting both internal and external coordination. As expected, we observed that the core-attention team launch increased internal coordination relative to both of the other conditions; specifically, the control condition yielded less internal coordination (Table 3, Model 3; $B = -1.09$, $p = 0.002$), as did the periphery-attention team launch ($B = -0.64$, $p = 0.002$). This effect remains even when controlling for core team members' experience working together, which was also positively associated with internal coordination ($B = 1.15$, $p = 0.006$), consistent with extant research. We also confirmed our expectation that the periphery-attention team launch increased external coordination relative to both the control condition (Table 3, Model 6; $B = -1.03$, $p = 0.019$) and the core-attention team launch ($B = 0.72$, $p = 0.006$). We further observed that the effect of the intervention remained significant when controlling for the senior resident's experience, which was also

Table 2. Descriptive Statistics and Correlations

Statistic	Mean	SD	Correlations										
			1	2	3	4	5	6	7	8	9	10	11
1 Average ALOS (days)	1.33	0.41											
2 Morning discharges (%)	0.37	0.15	0.06										
3 Average individual learning	6.26	0.36	-0.27	0.19									
4 Internal coordination	6.25	0.72	0.03	0.00	0.21								
5 External coordination	0.00	0.93	0.17	0.36	0.02	-0.02							
6 Core size (# members)	6.75	1.07	0.07	-0.11	-0.04	-0.03	0.14						
7 Attending experience (years)	15.95	11.30	0.04	0.11	0.11	0.04	0.09	0.18					
8 Senior resident experience (weeks)	2.24	1.89	0.01	0.10	-0.03	0.12	0.24	0.08	0.04				
9 Average intern experience (weeks)	2.69	1.22	0.04	0.18	0.11	-0.03	0.12	-0.42	-0.02	0.03			
10 Core experience together (average dyadic weeks)	1.19	0.17	0.12	0.12	-0.03	0.21	0.06	0.04	0.03	0.16	0.08		
11 Patient load (admitted patients)	14.34	5.83	-0.39	0.23	0.08	0.11	0.16	-0.01	0.03	0.20	0.03	0.01	
12 Average case severity (average APR DRG)	4.02	0.77	0.31	-0.15	-0.20	-0.01	0.22	0.13	0.05	0.09	0.13	-0.06	-0.14

Notes. Values in bold are significant at $p < 0.05$. $N = 91$. SD, standard deviation.

Table 3. Results from Multiple Membership Models Predicting Internal and External Coordination

	Internal coordination			External coordination		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Intercept	6.243*** (0.075)	5.272*** (0.825)	6.814*** (0.900)	0.000 (0.096)	−2.841** (1.036)	−1.412 (1.175)
Controls						
Week		0.019 [†] (0.010)	−0.023 (0.020)		0.006 (0.013)	−0.031 (0.026)
Average case severity		0.0003 (0.098)	0.068 (0.093)		0.209 [†] (0.123)	0.133 (0.119)
Patient load		0.006 (0.013)	−0.001 (0.012)		0.021 (0.016)	0.021 (0.016)
Core size		−0.020 (0.079)	−0.126 [†] (0.068)		0.155 (0.101)	0.166 [†] (0.099)
Core experience together		0.771 [†] (0.431)	1.146** (0.416)		0.053 (0.542)	−0.009 (0.527)
Attending experience		0.002 (0.006)	0.005 (0.006)		0.003 (0.008)	0.001 (0.008)
Senior resident experience		0.003 (0.042)	−0.024 (0.040)		0.075 (0.052)	0.106* (0.051)
Average intern experience		−0.081 (0.068)	−0.118 [†] (0.065)		0.107 (0.086)	0.143 [†] (0.083)
Key variables						
Control condition ^{a,b}			−1.088** (0.353)			−0.715** (0.258)
Periphery-attention team launch ^a			−0.638** (0.204)			
Core-attention team launch ^b						−1.030* (0.439)
Random effect variance components						
Core members	0.413 (0.154)	0.248 (0.286)	0.369 (0.077)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Residual	0.096 (0.135)	0.206 (0.280)	0.033 (0.048)	0.847 (0.126)	0.719 (0.107)	0.643 (0.095)
DIC	195.7	186.1	173.6	243.1	228.3	218.1
Observations	91	91	91	91	91	91

Notes. Standard errors are shown in parentheses. DIC, deviance information criterion.

^aIn Models 1–3, the core-attention team launch condition is the referent, given that it was expected to differ from the other two conditions with respect to internal coordination.

^bIn Models 4–6, the periphery-attention team launch condition is the referent, given that it was expected to differ from the other two conditions with respect to external coordination.

[†] $p < 0.10$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

positively associated with external coordination ($B = 0.106$, $p = 0.038$).

Individual Learning

To test Hypothesis 1, we evaluated the relationship between internal coordination and individual learning (see Table 4, Model 3). Because we measured learning at the individual level, we tested our prediction using a hierarchical linear model to account for the nesting of observations into individuals and teams. In support of Hypothesis 1, internal coordination was significantly and positively associated with individual learning ($B = 0.17$, $p = 0.004$). External coordination was also positively associated with individual learning, but the effect

size falls just below the standard threshold for statistical significance ($B = 0.080$, $p = 0.072$). Further, the relationship between internal coordination and learning was qualified by an interaction between internal and external coordination ($B = 0.10$, $p = 0.046$).⁷ Probing this interaction (see Figure 3), we observed that learning was greatest when the core team exhibited high levels of internal *and* external coordination. Plotting the marginal effects of external coordination on learning (see Figure 4) further demonstrated that, in contrast to the clear main effect that internal coordination had on learning, external coordination had a positive effect on learning only at higher levels of internal coordination. In addition, in the full model with controls included, individual learning is

Table 4. Results from Hierarchical Linear Models Predicting Individual Learning

	Dependent variable = <i>Individual learning</i>		
	Model 1	Model 2	Model 3
Intercept	6.421*** (0.444)	6.716*** (0.443)	6.631*** (0.440)
Key variables			
Core-attention team launch ^a	−0.231 (0.217)	−0.376 [†] (0.217)	−0.452* (0.218)
Periphery-attention team launch ^a	−0.308 (0.214)	−0.430* (0.213)	−0.466* (0.212)
Internal coordination		0.131* (0.056)	0.172** (0.059)
External coordination		0.082 [†] (0.045)	0.080 [†] (0.044)
Internal × external coordination			0.101* (0.051)
Controls			
Week	0.019 (0.012)	0.024* (0.012)	0.025* (0.012)
Average case severity	−0.106* (0.053)	−0.123* (0.051)	−0.121* (0.051)
Patient load	−0.001 (0.007)	−0.003 (0.007)	0.000 (0.007)
Core size	0.035 (0.044)	0.032 (0.043)	0.034 (0.043)
Core experience working together	−0.244 (0.223)	−0.380 [†] (0.221)	−0.398 [†] (0.219)
Attending experience	0.003 (0.003)	0.003 (0.003)	0.003 (0.003)
Senior resident experience	−0.006 (0.023)	−0.009 (0.023)	−0.006 (0.023)
Average intern experience	0.052 (0.038)	0.055 (0.039)	0.073 [†] (0.039)
Random effect variance components			
Individuals	0.190 (0.436)	0.195 (0.441)	0.191 (0.438)
Teams	0.032 (0.178)	0.024 (0.154)	0.022 (0.147)
Residual	0.351 (0.592)	0.348 (0.590)	0.348 (0.590)
Observations	477	477	477

Notes. Observations are nested in 210 individuals and 91 teams. Standard errors are shown in parentheses.

^aThe control condition is the referent.

[†] $p < 0.10$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

higher in later weeks of observation within our data collection window ($B = 0.02, p = 0.035$), and the average case severity of a core team's patients was negatively associated with individual learning ($B = -0.12, p = 0.018$).

Productivity

We tested Hypothesis 2, regarding the effect of external coordination on productivity, using multiple membership models and examining two measures of team productivity. One is the percentage of patients discharged between 6:00 a.m. and 11:00 a.m. (i.e., morning

discharges), a measure that reflects the near-term impact of a team whose members' involvement is essential to placing and executing discharge orders. Consistent with Hypothesis 2, we found that external coordination was associated with more morning discharges ($B = 0.05, p = 0.001$). Translating this effect, a one-standard-deviation increase in external coordination (0.93) was associated with a 5% increase in discharges completed in the morning. That is, for every 20 patients, one more patient would be discharged in the morning. The second measure of team productivity we used to test this

Table 5. Results from Multiple Membership Models Predicting Productivity

	Morning discharges				Average adjusted length of stay			
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Intercept	0.374*** (0.015)	0.357* (0.165)	0.481** (0.162)	0.053 (0.307)	1.325*** (0.043)	0.733 [†] (0.424)	0.855 (0.437)	2.654*** (0.805)
Controls								
Week		−0.007 [†] (0.004)	−0.006 (0.004)	−0.007 [†] (0.004)		0.004 (0.011)	0.006 (0.011)	0.012 (0.011)
Average case severity		−0.036 [†] (0.020)	−0.043* (0.019)	−0.035 [†] (0.019)		0.098 [†] (0.050)	0.089 (0.050)	0.055 (0.050)
Patient load		0.005 [†] (0.003)	0.004 (0.002)	0.004 (0.002)		−0.030*** (0.007)	−0.031*** (0.007)	−0.031*** (0.006)
Core size		−0.008 (0.016)	−0.018 (0.016)	−0.020 (0.016)		0.041 (0.041)	0.039 (0.042)	0.049 (0.041)
Core experience working together		0.076 (0.087)	0.083 (0.085)	0.105 (0.085)		0.191 (0.223)	0.158 (0.229)	0.070 (0.223)
Attending experience		0.001 (0.001)	0.001 (0.001)	0.001 (0.001)		0.000 (0.003)	0.000 (0.003)	0.001 (0.003)
Senior resident experience		0.008 (0.008)	0.002 (0.008)	0.003 (0.008)		0.003 (0.022)	−0.002 (0.022)	−0.005 (0.022)
Average intern experience		0.024 [†] (0.014)	0.016 (0.014)	0.012 (0.013)		−0.002 (0.035)	−0.005 (0.036)	0.008 (0.035)
Key variables								
Core-attention team launch ^a		0.088 (0.075)	0.078 (0.074)	0.101 (0.074)		0.029 (0.192)	−0.019 (0.199)	−0.115 (0.195)
Periphery-attention team launch ^a		0.165* (0.072)	0.113 (0.071)	0.137 [†] (0.072)		0.250 (0.185)	0.189 (0.190)	0.085 (0.188)
Internal coordination			−0.007 (0.021)	−0.015 (0.021)			0.031 (0.056)	0.066 (0.055)
External coordination			0.053** (0.016)	0.050** (0.016)			0.047 (0.044)	0.058 (0.042)
Average individual learning				0.065 (0.040)				−0.275* (0.104)
Random effects								
Core members	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.016 (0.167)	0.046 (0.090)	0.028 (0.103)	0.011 (0.103)
Residual	0.022 (0.003)	0.018 (0.003)	0.016 (0.002)	0.015 (0.002)	0.153 (0.168)	0.068 (0.089)	0.084 (0.103)	0.094 (0.104)
DIC	−90.7	−109.9	−119.8	−122.5	96.4	61.1	59.7	52.9
Observations	91	91	91	91	91	91	91	91

Notes. Standard errors are shown in parentheses. DIC, deviance information criterion.

^aThe control condition is the referent.

[†] $p < 0.10$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

prediction was the average length of time a team's patients stayed in the hospital controlling for their level of illness—or average severity-adjusted length of stay—which is a somewhat more diffuse measure, as it unfolds over longer periods and is affected not only by the team's work but also by factors such as organizational strain. When examining the effect of external coordination on average ALOS, we did not observe a significant relationship ($B = 0.05$, $p = 0.285$). These findings, together, offered some support for our prediction (Hypothesis 2) that external coordination improves productivity. We also found that the team's patients'

average case severity was associated with fewer morning discharges ($B = -0.04$, $p = 0.024$), and having a higher patient load was associated with an improved average ALOS ($B = -0.03$, $p < 0.001$).⁸

Exploratory Analyses

As noted in the theoretical background section, in addition to testing our predicted relationships, we also conducted exploratory analyses to further examine the dynamics of internal and external coordination and their combined implications for learning and productivity.

Table 6. Results from Multiple Membership Models Predicting Above-Mean Scores on Both Internal and External Coordination

	Model 1	Model 2
Intercept	−0.239*** (0.068)	−0.307 (0.572)
Conditions		
Core-attention team launch ^a	0.283* (0.117)	0.721** (0.246)
Periphery-attention team launch ^a	0.125 (0.119)	0.516* (0.238)
Controls		
Week		−0.029* (0.014)
Average case severity		0.077 (0.065)
Patient load		−0.008 (0.009)
Core size		−0.017 (0.054)
Core experience working together		0.512 [†] (0.285)
Attending experience		0.002 (0.004)
Senior resident experience		0.029 (0.028)
Average intern experience		−0.014 (0.045)
Random effect variance components		
Core members	0.000 (0.000)	0.000 (0.000)
Residual	0.211 (0.031)	0.188 (0.028)
DIC	116.7	106.4
Observations	91	91

Notes. Standard errors are in parentheses. DIC, deviance information criterion.

^aThe control condition is the referent.

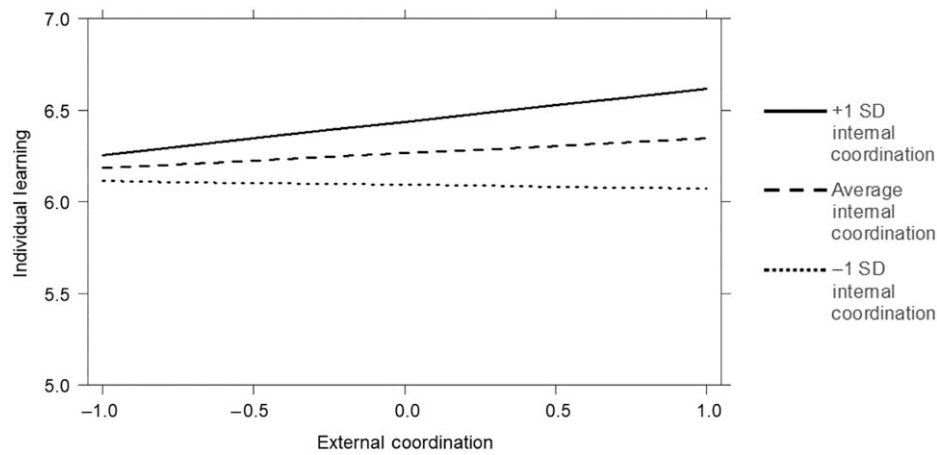
[†] $p < 0.10$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Learning and Productivity. The expected relationship between individual learning and team productivity was not obvious based on extant research; thus, we conducted a post hoc exploratory analysis to examine if there is evidence of synergistic gains in this context or if these dynamic teams generally cannot accomplish both successfully. Because productivity is a team-level variable, we aggregated our measure of learning at the team level by taking the average learning of the trainees on each core team. As shown in Table 5, Models 3 and 6, we observed a positive relationship between learning and both of our team productivity measures; for morning discharges, the effect of learning was not significant ($B = 0.06$, $p = 0.102$; with a 95% confidence interval (CI) of $[-0.01, 0.14]$), but learning did have a significant relationship with average ALOS ($B = -0.27$, $p = 0.008$; with a 95% CI of $[-0.48, -0.07]$). In other words, a one-

standard-deviation increase in a team's average individual learning (0.36) was associated with an 8% decrease in the average ALOS for that team's patients.

Indirect Effects. We further explored the fit of our data with a model capturing the full sequence of theorized relationships—how a team launch increasing internal versus external coordination impacts learning and productivity—by estimating a structural equation model with maximum-likelihood estimators (Bollen 2005). We used the lavaan package in R and implemented Bollen-Stine's model-based bootstrapping (5,000 samples) to test statistical significance (Rosseel 2012). We again conducted the analysis at the team level, requiring that we use an aggregated learning measure, and we again take the average of the individual learning scores for each team. Given the resulting reduction in observations when predicting learning, to increase the statistical power of our analysis, we also exclude control variables, yielding a ratio of sample size to number of free parameters of over 6:1 (i.e., greater than a recommendation of 5:1; Kenny 2020). In addition to these adjustments, another caveat to keep in mind while reviewing these exploratory results relates to the ways our data violate an important assumption of this model in that our units of analysis are not independent. Therefore, we emphasize the exploratory nature of this analysis and view the results as suggestive for future work that could ideally test these relationships in a data set that would enable more robust analyses.

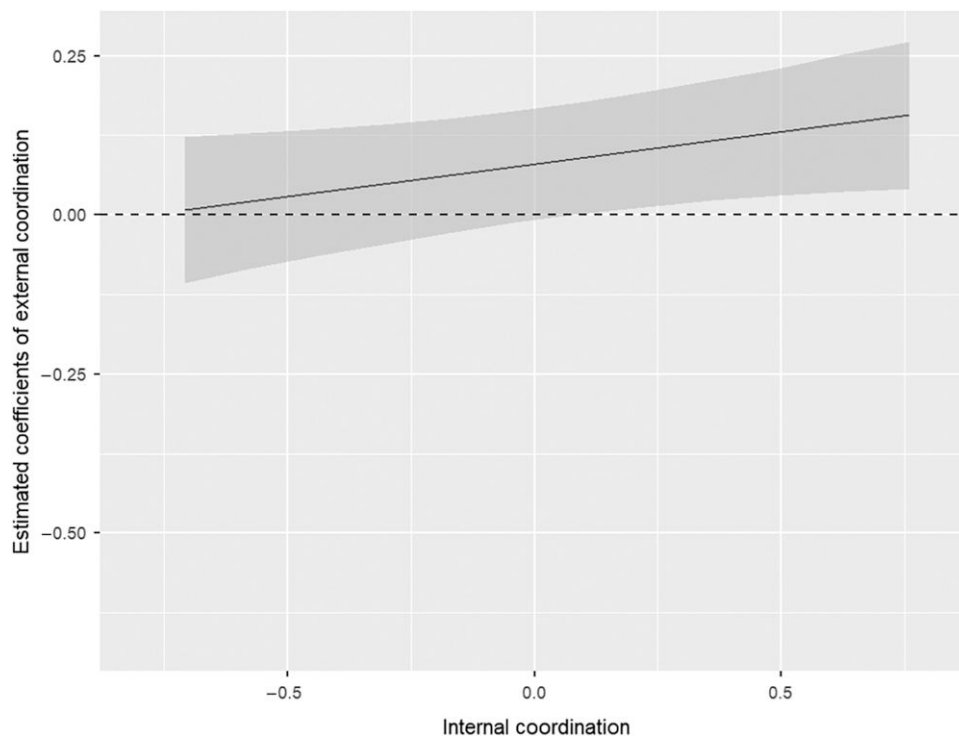
With these cautionary notes in mind, we first observe that, as shown in Figure 5, the fit statistics indicate that the model is a reasonable fit for the data: ($\chi^2(14) = 26.25$, $p = 0.566$; comparative fit index (CFI) = 0.796; root mean square error of approximation (RMSEA) = 0.098; standardized root mean squared residual (SRMR) = 0.088).⁹ With regard to learning, as shown in Table 7, the analysis suggests that the core-attention team launch indirectly and positively impacted learning via internal coordination, conditional on external coordination. Specifically, relative to the control condition, when external coordination was one standard deviation above average, the core-attention team launch had a positive impact on learning ($B = 0.13$, $p = 0.022$). Relative to the periphery-attention team launch, when external learning was one standard deviation above average, there was also a positive effect, though it did not reach standard thresholds of statistical significance ($B = 0.10$, $p = 0.078$). With regard to morning discharges, we observed that the periphery-attention team launch indirectly improved the number of morning discharges via external coordination relative to the control condition ($B = 0.04$, $p = 0.015$) and the core-attention team launch ($B = 0.03$, $p = 0.048$). Finally, the coefficients for the indirect effect of the core-attention team launch on average ALOS via internal coordination

Figure 3. Interaction Effect of External and Internal Coordination on Individual Learning

Note. For ease of interpretation, we use a simplified ordinary least squares model to visualize the interaction.

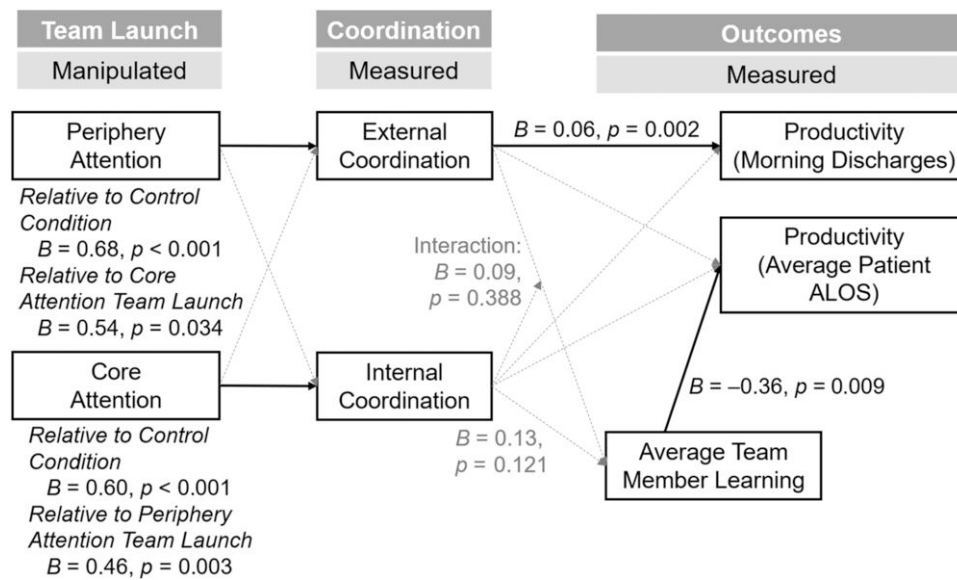
and learning, conditional on external coordination, were in the predicted direction and larger than the coefficients for this relationship in the control condition and periphery-attention condition (when external coordination was one standard deviation above average), but were not statistically significant ($B = -0.05$, $p = 0.108$ and $B = -0.04$, $p = 0.188$, respectively).

Of note, these patterns of (conditional) indirect effects exist despite a component of those pathways (the effects on learning) being nonsignificant in the model. We suspect that the weaker effects on learning, and on average ALOS via learning, are due in part to the forced aggregation of the individual learning measure to the group level—a constraint that implies reduced

Figure 4. Estimated Coefficient of External Coordination on Individual Learning by Internal Coordination

Notes. For ease of interpretation, we use a simplified ordinary least squares model to estimate that the effect of external coordination on learning (y -axis) is positive when the mean-centered internal coordination variable is above 0.12. The 95% confidence intervals are depicted in the shaded gray region.

Figure 5. Structural Equation Model with Key Study Variables



Notes. In this figure, $\chi^2(14) = 26.25, p = 0.571$, CFI = 0.80, RMSEA = 0.10, and SRMR = 0.09. Direct effects represented by dashed gray lines are not significant.

power of our analysis and that washes out the variance in individual scores within a team.

Achieving Both Internal and External Coordination. A final exploratory analysis focused on probing the observation in our data that learning, and thus productivity, was greatest among teams scoring high on the measures of both internal and external coordination, leading us to wonder what antecedents enable a team to achieve high levels of both. Starting with a fairly coarse approach, we identified teams that scored above average on *both* internal and external coordination (or “high–high” coordination teams) and compared rates across study conditions. In doing so, we found that the core-attention team launch generated the most high–high teams (52%, or 12/23 teams), followed by the periphery-attention team launch (36%, or 8/22 teams), followed by the control condition (24%, or 11/46 teams). Using multiple membership models, and including study control variables,

we applied dummy codes to the coordination categorizations, coding “high–high” as one (and zero otherwise), and observed that both the core-attention team launch and the periphery-attention team launch conditions were more likely to produce high–high teams relative to the control condition ($B = 0.72, p = 0.003$; $B = 0.52, p = 0.030$, respectively; see Table 6, Model 2). We take these exploratory findings to suggest that both team launches may have the potential for spillover effects, altering not only the intended coordination behaviors, but possibly prompting teams to consider both areas of coordination. We also note, even more speculatively, that the patterns suggest that more spillover occurs across internal and external coordination in teams receiving the core-attention team launch, though given the small numbers of teams, we reinforce again the very preliminary nature of these observations and hope future work can build on them to design and robustly test predictions related to these effects.

Table 7. Summary of (Conditional) Indirect Effects

Mediation	Effect	p-value
Indirect effect		
Periphery-attention team launch → external coordination → morning discharges	Relative to the control condition: 0.04	0.015
	Relative to the core-attention condition: 0.03	0.048
Conditional indirect effects (external learning = one standard deviation above average)		
Core-attention team launch → internal coordination → learning	Relative to the control condition: 0.13	0.022
Core-attention team launch → internal coordination → learning	Relative to the periphery-attention team launch: 0.10	0.078
Core-attention team launch → internal coordination → learning	Relative to the control condition: −0.05	0.108
Core-attention team launch → internal coordination → learning → average ALOS	Relative to the periphery-attention team launch: −0.04	0.188

Discussion

We demonstrate that in dynamic teams, brief team launches implemented with only a team's core members can enable more effective team coordination. Specifically, we find that directing attention to core members' roles and workflows enhances internal coordination, while directing attention to potential interdependencies with periphery members who are external to the core group enhances external coordination. Further, we find that although external coordination fosters one measure of productivity in the setting of hospital inpatient teams—the percentage of patients discharged in the morning—it is the combination of internal *and* external coordination that facilitates core members' learning, which is associated with overall team productivity: average adjusted patient length of stay. In conducting this field experiment and reporting our findings, we make a number of theoretical contributions to the existing literature, particularly in reference to learning, dynamic teams, and team launches.

Revisiting the Learning–Productivity Trade-Off

Whereas some research might suggest a trade-off between learning and productivity (Sobrero and Roberts 2001, Singer and Edmondson 2008, Dennis et al. 2014, Rabinowitz et al. 2016), other work suggests the potential for synergistic gains (e.g., see Edmondson et al. 2007, Wiese et al. 2022). Reflecting on the observed relationship between coordination and learning in our study, as well as the different results we observed for the relationship between learning and our two productivity measures (morning discharges and average ALOS), suggests multiple theoretical implications for consideration in future research on this topic.

Although we initially predicted that internal coordination would be important for learning, we additionally find that internal coordination and external coordination jointly explain variance in our learning measure. Thus, although we predicted that core members can learn through interaction with fellow core members, it appears that core members—when they engage in internal coordination—can also learn from the inputs of periphery members obtained through external coordination. These findings are consistent with some extant work at the team level documenting that when a team engages in a high level of both internal and external learning activities—and particularly when the internal activities reflect a more reciprocally interdependent pattern of work—the team (theoretically, at least in part because those activities led to more individual learning) performs best (Bresman 2010, Myers 2021; cf. Wong 2004). Moreover, our theorizing complements that of Myers (2021) in that we suggest our pattern of results underscores the importance of creating conditions that

enhance absorptive capacity—here, among the individuals working in dynamic team settings. Our initial focus when predicting learning was the idea that interactions within the core team—comprising individuals with similar training and tasks—would afford visibility of other's work and opportunities to learn from each other that are often absent in such dynamic settings (Myers 2018, 2022). In sum, our findings further suggest that when a core team's members work together, they also boost each individual's absorptive capacity for integrating outside information (e.g., Venkataramani and Tang 2024) such that they can learn from interactions with individuals who offer unique insights and often span professional and disciplinary boundaries—indeed, this underlying absorptive capacity gain could explain why, when the foundation of internal coordination was present, external coordination with periphery members also boosted learning.

In further probing how individual learning relates to productivity, a deeper consideration of the different productivity measures in our study suggests additional insights. Although both indicators of productivity were thought to benefit from enacted interdependence between the core team and periphery members, the nature of each outcome may mean that the form of interdependence that could support each varied. We first consider the process of discharging patients in the morning. As the day starts, it is likely that there is a clear plan in place. That is, certain criteria, if met, would mean the patient could be discharged. The matter of determining whether those criteria are met and then ensuring the critical steps are in place for discharge likely involves the periphery (e.g., the core team may need final input from consulting specialties in order to place the discharge order), but it likely does not necessitate a high level of internal coordination. Thus, it is logical that the quality of external coordination would directly impact the efficiency of execution of a task such as this. By contrast, reducing overall patient length of stay involves a variety of interrelated decisions and activities initiated very early in the treatment process that likely benefit from higher levels of reciprocal interdependence among core team members in addition to engagement with periphery members; namely, the team must gather information, synthesize it, and make decisions, and then delegate and execute (Grippa et al. 2018). In moving through these activities, a qualitative study of inpatient teams suggests that when teams engage in external coordination to gather information and internal coordination to delegate, a synergy is created that enables collective synthesis and decision making—involving both core and periphery members (Mayo 2022). That step of reasoning through the decisions collectively or hearing how others do so can offer more insights that could be adapted and applied to future, related events—in other words, more individual

learning. This could explain why a measure of patients' average adjusted length of stay benefits from both external coordination as well as internal coordination, in that the two ensure that relevant information is integrated and that delegation is clear, but additionally because they jointly create opportunities for accessing new knowledge while also making sense of its relevance, such that members gain both knowledge and an ability to integrate and apply it (Myers 2018).

Finally, we note that our multilevel approach, an approach that has been called for in the literature more broadly (House et al. 1995, Hackman 2003), allows us to identify a positive relationship between learning assessed at the individual level, and productivity assessed at the team level in terms of average ALOS. Our work thus adds to research on the tensions and synergies between individual and unit-level outcomes (Kerrissey et al. 2023), and echoes adjacent research suggesting that team-level learning can support organizational-level productivity (Harvey et al. 2022).

Dynamic Teaming, Right from the Start

Dynamic Teaming. As reported previously, we have observed a dramatic increase in the use of dynamic teams in many industries, with members working together briefly and on multiple teams at once, and with fluid, dispersed team membership (Lee and Edmondson 2017, Mortensen and Haas 2018). These dynamics strain members' attention, making it difficult to know where to focus and with whom to work (Mayo 2022). A burgeoning set of theory is emerging on these more dynamic teams (Humphrey and Aime 2014, Bernstein et al. 2018, Mortensen and Haas 2018, Mayo 2022), and our randomized controlled field experiment advances both an understanding of the processes that contribute to a dynamic team's effectiveness and the causal impact of team launches—implemented with only a team's core members—on coordination.

Our choice to disentangle internal coordination and external coordination allowed us to account for the distinct effects of each behavior. And by disentangling these behaviors from each other *and* from learning, we are able to integrate work that has tended to focus on either internal and external learning activities (Bresman 2010, Bresman and Zellmer-Bruhn 2013, Myers 2021) or internal and external coordination (Choi 2002, Mayo 2022), creating a path forward for understanding how these constructs relate. Given our observation that internal and external coordination jointly explain variance in our learning measure, which in turn explains variance in our broader measure of productivity, we suggest that our findings offer insight into the effective dynamic “teaming” practices (Edmondson 2012) that organizations should seek to cultivate.

Our work also speaks to how teams might cultivate those teaming practices by focusing on the content of

what happens in a team “right from the start” (Ericksen and Dyer 2004, p. 469) and by teasing out the implications of directing attention to the core versus the periphery. Although each intervention tested in our setting had a distinct effect—either on internal or external coordination—our exploratory analyses suggested that interventions on the core or the periphery could both spill over and affect the ability of a team to achieve not just high levels of internal coordination but also external coordination, or vice versa. Moreover, we find some suggestive evidence that the core intervention might have greater spillover effect relative to the periphery intervention. As in our results section, we again caution against overinterpretation of these findings, and we hope our results will spark additional research to both replicate and extend their implications for practice. Given the absence of both expansive time to plan together and the ability to identify who should be involved a priori, it is critical to identify the essential conditions to put in place to best facilitate dynamic team functioning. One possible explanation for our findings is that our periphery-attention intervention directed focus on and enabled external coordination, but may have simultaneously resulted in more siloing of work among core team members—for example, allocating specific members to interface with different patients—in a manner that inhibits internal coordination. In contrast, the core-attention intervention directed focus on and enabled more interdependent work among the core team members, and this may have also increased absorptive capacity. In this way, the core-attention intervention not only facilitated internal coordination, but also enabled the team to better capitalize on periphery members' knowledge and resources both for individual learning and for collective decision making, ultimately resulting in better performance (as gauged by ALOS). The possibility of synergistic gains between internal and external coordination is consistent with prior qualitative work on inpatient teams, as noted above (Mayo 2022). Yet future work is needed to explore these ideas and further develop our understanding of what causally affects dynamic team functioning, learning, and performance. Our choice to test team launches focused on either the core or the periphery allowed us to disentangle some of the effects of internal versus external coordination, a key goal of our work. Future work could look at the benefits of team launches with different mixes and levels of focus on internal and external coordination to see if there are opportunities to amplify potential synergies.

Team Launches and Huddles. While advancing theory and empirical evidence related to dynamic teams, our work also extends and adapts theory about the target and content of team starts. Work on team launches—whether generally acknowledging that “long-lasting effects flow from events early in the life of a group”

(Hackman 1987, p. 336; see also Mathieu and Rapp 2009) or explicitly characterizing a team launch (Kerrissey and Singer 2020), team start (Ericksen and Dyer 2004, Woolley 2009), or brief that occurs prior to work beginning (e.g., “premission briefings” (Marks et al. 2000, p. 973) or “prebriefs” (Traylor et al. 2021, p. 6))—highlights how gathering team members at the start of the team’s lifespan offers an opportunity to set the stage for better team processes and performance. However, dynamic teams tend to have a core team, often due to larger staffing decisions (Kerrissey and Singer 2020, Mayo 2022), that is supported by more temporary, fluctuating periphery contributors. Because the periphery shifts as tasks demand, the entire team cannot be known at the team’s outset, such that it is impossible to gather the full team for a team launch. This is quite different than the underlying assumptions of extant work on team launches, which presumes the presence of the entire team at a team launch. We thus integrate work on the importance of “strategic core roles” (Humphrey et al. 2009, p. 51) to intervene solely with the core members. Additionally, this allows us to disentangle the role of content focused on work within the team’s core, as opposed to work with periphery contributors. To do this, we break from past research that views team launches as a chance to establish clear team roles and a clear team boundary (Hackman 2011), to instead test the implication of establishing scaffolding for flexible core roles and a shifting team boundary.

Related work on team huddles, a practice commonly used in healthcare settings, reveals that although the term “huddle” can refer to a variety of activities (for a review, see Franklin et al. 2020) and scholars have noted the lack of a standard definition (Rodriguez et al. 2015), there are features common across studies that are similar to those of team launches. First, like launches, huddles serve as a tool for gathering people and directing attention at the start of a day or shift (Shaikh 2020). As with team launches, our work makes a few contributions to the research on huddles. First, like team launches, huddles are also commonly used within more stable, bounded teams, enabling all of a team’s members to participate (Jain et al. 2015, Rodriguez et al. 2015). Even when huddles span hospital divisions, they might focus on a defined “interprofessional team,” one that consistently includes the same roles from across divisions (McBeth et al. 2017). Thus, here, too, our work suggests expanding the focus of huddles to consider an application to dynamic teams that incorporates consideration of periphery members who cannot be present. Second, although the focus of huddles can vary, we can apply a well-adopted distinction between taskwork and teamwork (Mathieu and Rapp 2009) to better understand them; where the team launch work reviewed above (and the content in our core-attention team launches) typically focuses on teamwork, huddle content has tended to focus on taskwork (e.g., reviewing

patient care plans (Alaraj et al. 2017), alarm settings (Bonafide et al. 2018), administrative or clinical challenges (Pannick et al. 2017), or safety risks (Shaikh 2020)). Although some prior work on team launches has focused on both taskwork and teamwork (Mathieu and Rapp 2009), and this approach would likely be useful in any future setting, our findings suggest that future work on huddles in healthcare would do well to expand the focus to include *teamwork*. Third, whereas work on both team launches and huddles commonly examines their impact on team/unit-level or organization-level outcomes (Rowan et al. 2022), we look at the effect of the team launches on *individual* learning. More work to explore cross-level effects of team activities on individual outcomes such as well-being or burnout, a pervasive challenge in many settings, could be fruitful. Finally, recent work highlights a lack of high-quality evidence regarding the efficacy of huddles (Franklin et al. 2020, Rowan et al. 2022); our research helps to answer that call.

Toward a Broader Understanding of Team Scaffolds. Our focus on team starts as a mechanism for supporting dynamic teams builds on and extends prior theory on team scaffolds (Valentine and Edmondson 2015). Specifically, we join with others who have highlighted the basic underlying function of managing attention in organizations (March and Simon 1958, Ocasio 1997, Bernstein et al. 2018) to suggest that attention may be a central mechanism underlying the efficacy of team scaffolds (Valentine and Edmondson 2015) as a substitute for formal organizational structures. We posit that there may be a broad class of scaffolds, including team launches like those studied here, that operate via directing attention and thereby guiding a team’s collective interactions. For instance, here we focused on team launches to enable internal and external team coordination, but other scaffolds could help individual members identify others’ expertise to aid the team in collectively integrating information (Woolley et al. 2008, Gupta and Woolley 2018) or to facilitate decision making (Chidambaram et al. 2021). This suggestion adds to the growing body of research that highlights collective attention as a fundamental process underlying collective intelligence (Riedl and Woolley 2017, Gupta and Woolley 2018, Mayo and Woolley 2021, Woolley et al. 2023). Future work to explore the underlying role of attention in team launches like those used in our setting, and to consider a broader class of team scaffolds that direct attention in dynamic teams, could further enable organizations’ ability to cultivate effective dynamic teams.

Boundary Conditions and Generalizability

Promoting External Coordination. In collecting data from periphery members, we ultimately focused on nurses, as theirs was the only external member role consistently involved in the patient care teams in our

setting. There are at least a couple of implications of this decision. The first relates to the effect of the intervention on external coordination. For every patient, a core team is assigned, and a nurse is also assigned. Nurse workstations are also proximal to the patient rooms, a space frequented by the core team members. This consistency and visibility could make nurses more top-of-mind for core team members relative to other positions (e.g., pharmacists, specialists, social work) that are more inconsistently involved in patient care and more removed geographically from the spaces a core team typically inhabits. This could lead our intervention effects on external intervention to be overestimated—the effect could be smaller where there is more inconsistency and dispersion. However, nurses are also commonly viewed as occupying a lower-status position than physicians (Bransby et al. 2023). This could lead our intervention effects to be underestimated—the effect might be larger when promoting interaction with higher-status individuals, to whom the core team might more readily turn. Future work is needed to explore the influence of periphery members' attributes further.

External Coordination and Productivity. The second implication of focusing on nurses relates to our analysis of the relationship between external coordination and productivity. As already noted, we assume in our study that external parties hold relevant information or capabilities, but, where this is not possible, it follows that external coordination would also be less important or possibly counterproductive (Woolley et al. 2013). Additionally, it is important to note the role of automated information systems in facilitating work in ways that minimize the need for the core team to actively do so. Given the expanding role of the electronic health record (EHR), there is at least a moderate level of interdependence between the core team and periphery members built into many healthcare settings. For instance, at the end of a patient's admission, when the core team places a discharge order into the hospital's EHR system, sequential interdependence that is hard coded into their EHR system enables a nurse to see the order and complete the next steps without the core team ever taking additional action to facilitate coordination. Similarly, when prescribing medicine or ordering tests, periphery members consulting the EHR can identify and execute their tasks. Theoretically, such a system should reduce or remove the need for extensive external coordination. However, despite substantial investments in EHR systems, some argue that the benefits for efficiency in coordination have been modest (DesRoches et al. 2010, Ayaad et al. 2019, Dutta and Hwang 2020) suggesting the continued need for team members to coordinate with each other directly, and not simply through the EHR system. That external coordination was valuable in our study's setting, where the EHR system is in regular

use, further confirms its importance. It also suggests that the importance of external coordination is likely even greater in other settings where some aspects of coordination are not as facilitated by information systems.

Learning in Dynamic Teams. In this setting, we examined individual learning of core members who were specifically in the role of trainees, where learning is an explicit goal. The amount of learning reported varied in our experiment, despite learning being a stated priority. Future research could focus on the degree to which the findings generalize to settings in which learning is less of an explicit goal, and whether facilitating internal and/or external coordination may further amplify the impact on individual learning.

Limitations and Future Research

We would be remiss if we did not acknowledge some limitations to our work that have implications for future research. First, we were able to account for the role of shared experience within the core team. Perhaps more critically for our study, this team characteristic did not significantly differ across our conditions, giving confidence that this does not explain any differences across our interventions. However, future work would likely benefit from also testing the role of core-periphery shared experience (e.g., for an analysis of attending/resident-nurse relations; see Kim et al. 2023).

A second limitation of this research setting is that core team members worked on different teams at different time points over the course of the study, which creates a lack of independence across the teams in our sample. We addressed this issue statistically using multiple membership models that partition the variance explained by team members. This solution will be helpful in future research on dynamic teams, where individuals work on multiple teams either simultaneously or over time during a study period. That said, we were limited in our ability to implement tests of indirect effects, as current models do not allow us to do so while accounting for multiple membership. We opted to implement a structural equation model and interpret it with caution. To that end, we acknowledge that the weak results in the predictions of learning could be driven by the violation of the assumption of independence or the loss of statistical power in aggregating learning to the group level to implement the test of learning on average ALOS. Taken together, although our results advance a multilevel perspective on the impact of these interventions across levels of analysis, additional advances in multilevel statistical modeling will be helpful to further refine our understanding.

Relatedly, whereas we worked closely with hospital administrators to limit "cross-contamination" of our treatments through ensuring that core team members were only scheduled on teams in blocks assigned to the

same condition, future work in similarly dynamic team environments might more specifically evaluate the diffusion of work practices that can result from a lack of independence which in some settings could be exploited as a feature rather than a bug.

Finally, there are a few limitations to our intervention design and our related tests of their effects that suggest avenues for future research. First, as noted above, we opted to compare interventions specifically focused on either the core or the periphery. This was driven by an interest in maximizing the differences between conditions to disentangle the effects of internal and external coordination, and was reinforced given the limited sample possible in our context, which precluded the additional test of a more nuanced mixture of core and periphery attention. We see this as a first step in building toward future work that could test a team launch that combines the two. Second, although some individuals received the treatment (a team launch) more than once in our study, the high turnover in our core teams due to member rotations through other areas of the hospital for training, or through other administrative or research roles, results in little variance in treatment exposure. Future work that could better unpack the effects of varying treatment exposures could also facilitate smarter intervention design. Finally, we are limited by the cross-sectional lens on teams in our study. Future work to explore the effects over time could be achieved via longitudinal work that explores how team processes unfold, for instance, examining how long-lasting the intervention effects are within a team's lifespan. Additionally, future work could step back to take a broader view of the unit in which teams work, exploring how behaviors could diffuse over time throughout the entire unit. Both could be insightful with respect to refining effective interventions. This could also have important implications when taking into account another limitation of this work: the self-report nature of the learning measure. Future work could build from our findings by instead, or also, using supervisor ratings of learning or a more objective test of learning, and this could be particularly informative regarding the relationship between learning and productivity if a longitudinal approach is adopted.

Conclusion

The results reported here demonstrate that even in dynamic teams, which are temporary and lack fixed and clear boundaries, the existence of core teams can be leveraged; team launches implemented with the core team members can enable coordination and the potential for both individual learning *and* productivity. In an era when increasing sums are spent on new technologies to enable coordination in complex systems, this work highlights the power of a simple and low-cost approach that builds on the foundational role of human

attention in the structure and organization of work. And if we are correct in assuming that the shift to more dynamic and less structured environments for team-work is a development that is unlikely to be reversed, then uncovering interventions such as this one could have considerable theoretical and practical importance.

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Endnotes

¹ To assess the study design's power, data were simulated based on anticipated relationships among the constructs of interest, as well as hypothetical individual effects (e.g., the effect of having a highly skilled attending physician), effects of time (e.g., over time, we could expect units to improve across the board with regard to some coordination behaviors), and noise for each construct of interest (i.e., error). One hundred data sets were simulated for a range of sample sizes. Planned analyses were then run with each simulated data set and used to recover the power to detect each hypothesized effect. The power for a range of sample sizes and each effect is provided in Online Appendix A. As described further in subsection Block Random Assignment, each team in the second phase received one of two team launches. In the first week that we implemented the team launches, we did not implement the team launch in one of the four core teams because of a last-minute scheduling change in which a replacement core team member was added who was scheduled to receive the alternate team launch in a later week. To avoid contamination, we did not implement a team launch for that core team and excluded it from analyses. Additionally, in the final week of the study, because of another last-minute scheduling change, a replacement core team member was slotted to receive one of the team launches, but this individual had already received the alternate team launch in earlier weeks. We excluded this core team from analyses. Finally, we included core teams in our analyses only if they received a minimum of two nurse ratings, leading us to exclude an additional three core teams. Subsequently, from the initial set of 96 teams, our analyses include 91 core teams (46 in the control condition, 23 in the core-attention team launch condition, 22 in the periphery-attention team launch condition).

² Team launches were observed by an experimenter or research assistant to ensure that they were implemented.

³ Results are consistent whether using the composite measure reported in our primary analyses, solely the percentage of nurses invited to rounds, or solely the two Likert-scale survey items about being included in decision making and having input.

⁴ Results are consistent in both direction and significance whether using the average learning score of a core team's trainees or the minimum. There was not enough variance in the team's maximum score for that to explain productivity. We also explored learning by role but did not observe evidence that either learning differed by role or that the relationship between learning and average ALOS was driven by a particular role.

⁵ A team's patient-related scores reflect all patients for which they provided care. About 10% of the patient sample was cared for by more than one team. Although the efficiency of treatment for patients overlapping multiple core teams is dependent on the coordination of each team involved, even when the analysis is repeated based solely on the first treating team, results are consistent in direction with those reported, though the reduction in observations included leads to wider confidence intervals and weaker results straddling thresholds for statistical significance.

⁶ Results do not differ whether including or excluding the average case severity of core teams' patients.

⁷ To facilitate interpretation, internal coordination was centered before exploring the interaction effect. External coordination was already centered because of its initial scoring method.

⁸ One possibility for the relationship between patient load (measured as admissions) and improved length of stay is that admissions tend to go up when respiratory illnesses spread in a community. As such, teams receiving more admissions may be receiving more admissions of the same type, and this could create some efficiencies relative to having a more varied set of patient case types. This logic would be consistent with past work demonstrating a relationship between the volume of certain cases and outcomes (e.g., see Kahn et al. 2006). Another is that high workloads stimulate teams to develop new and better ways of coordinating as a means of adapting to the increased volume, which can have spillover effects on improving performance in many different areas of their work (Joshi et al. 2016). Lastly, some studies have demonstrated that higher patient volume forces a higher density of interactions between doctors and more information sharing, which stimulates better coordination among them and facilitates follow up across different patient cases (Pollack et al. 2015).

⁹ These statistics can be more variable with smaller sample sizes, and RMSEA and SRMR tend to be inflated in models with lower degrees of freedom (Kenny 2020).

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